
Formation Damage During Drilling:

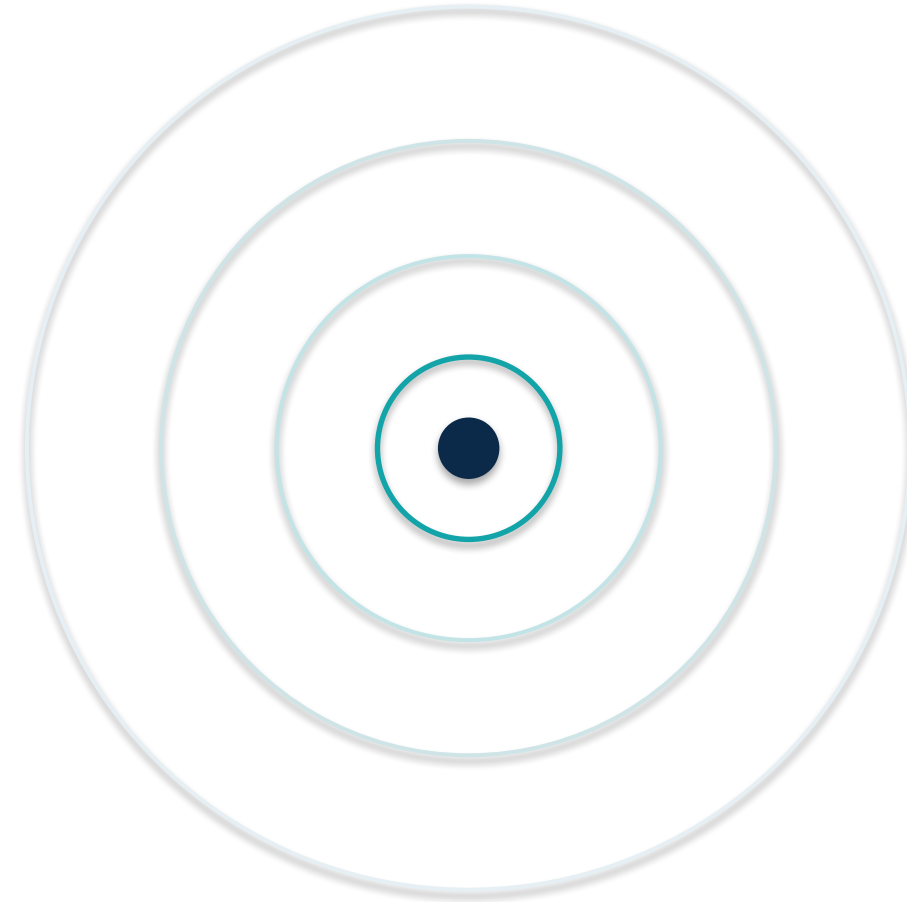
Mechanisms, Skin Quantification, and Mitigation Strategies

Supervisor: [Prof. Name]

Department of Petroleum and Gas Engineering · [University]

Seminar · [Month Year]

Image: [source / licence]



23 slides + 5 hidden backups

Same drawdown, but damage cuts the rate by more than half

-55%

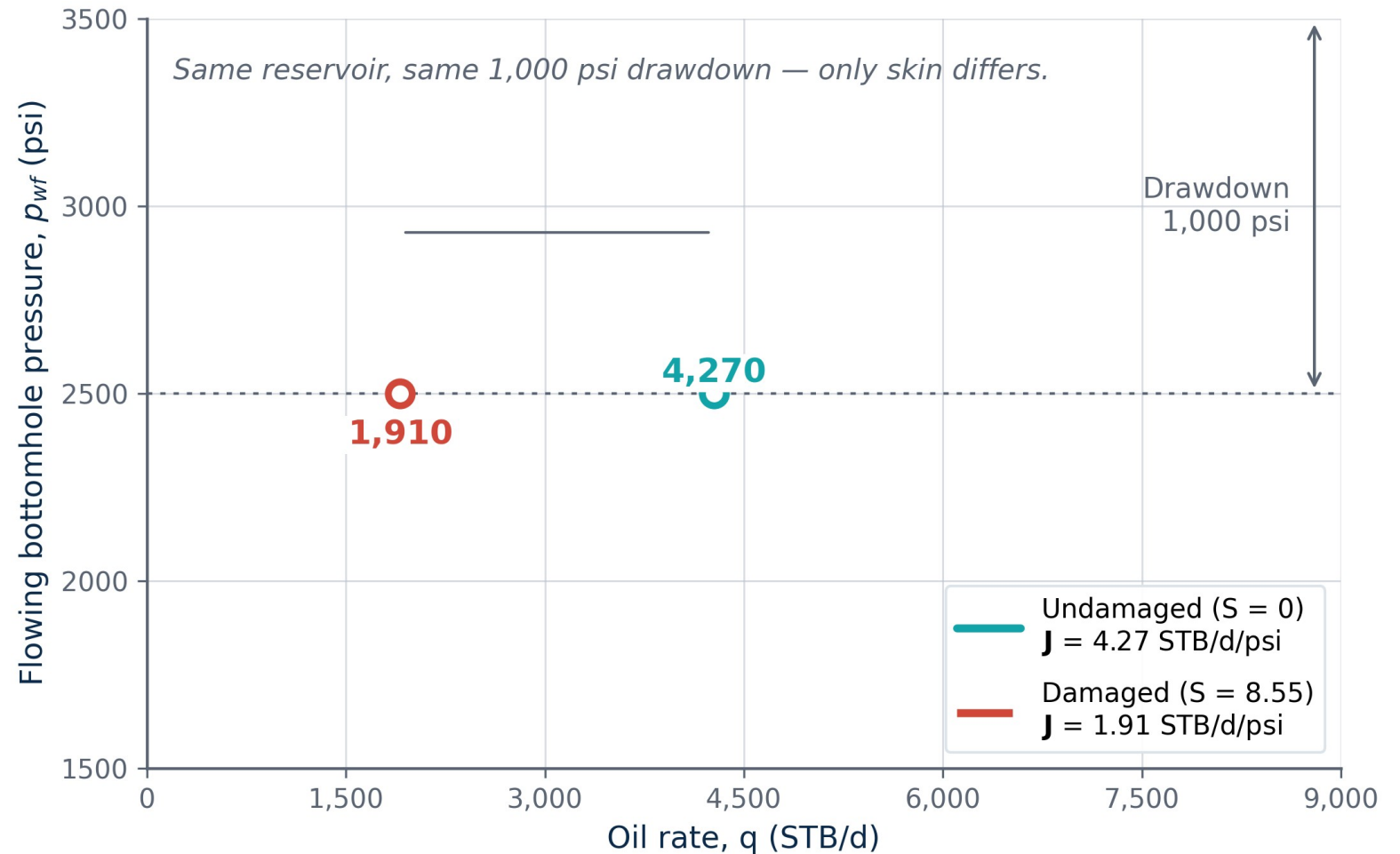
rate at the same drawdown

4,270 → 1,910

STB/d, undamaged → damaged

($S = 0 \rightarrow S = 8.55$)

At $p_{wf} = 2,500$ psi, same drawdown.



Damage is invisible while drilling but costly during production

Problem: drilling fluids impair near-wellbore permeability

1

1. Mechanisms

Explain how damage forms

2

2. Quantification

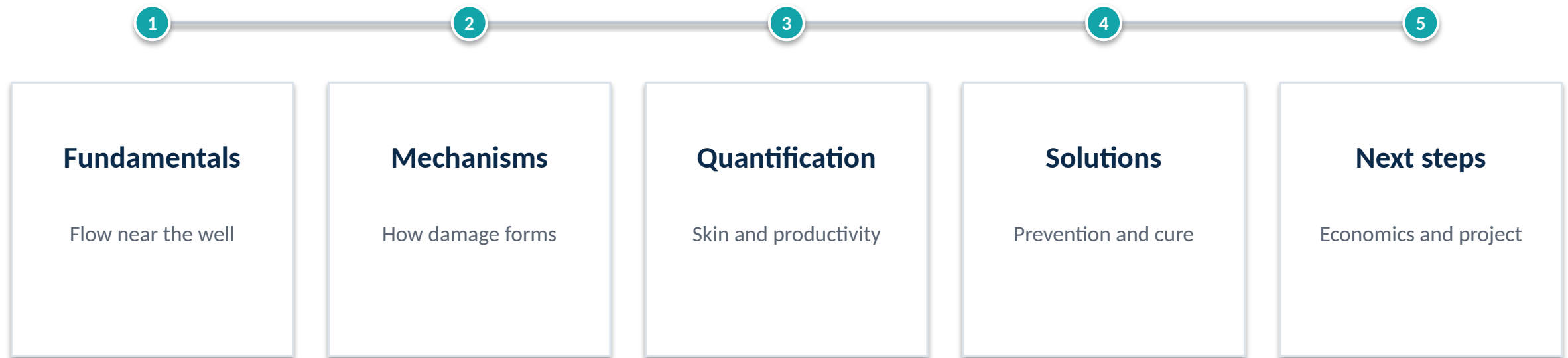
Measure the loss with skin

3

3. Strategies

Compare prevention and remediation

Five steps, from flow physics to my project plan



The near-wellbore zone controls inflow

Fundamentals

Mechanisms

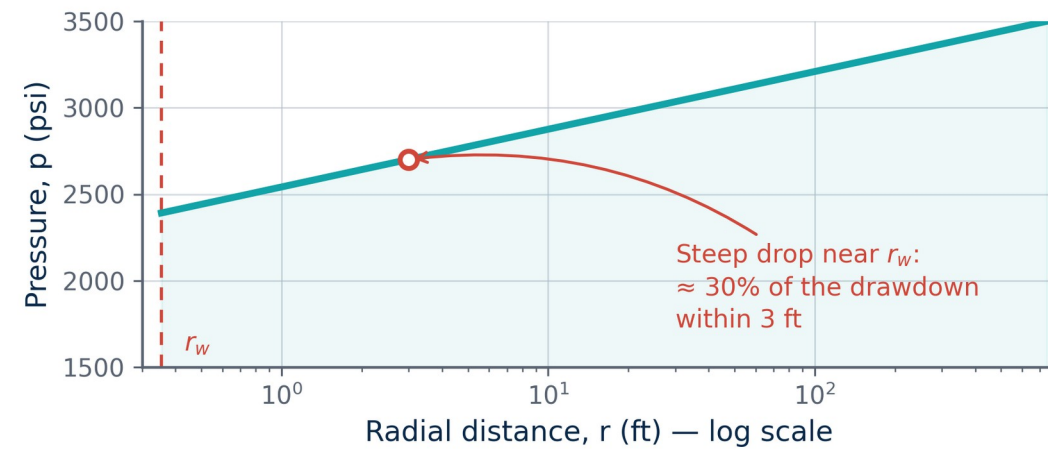
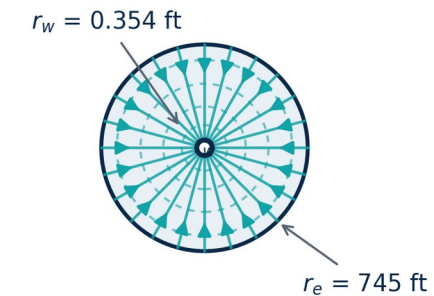
Quantification

Solutions

Next steps

- Flow area shrinks toward the well
- Pressure gradient is steepest near the wellbore
- About 30% of drawdown occurs within 3 ft
- Small damaged zone, large effect

Top view: radial flow into the well



Skin factor measures deviation from ideal radial flow

Fundamentals

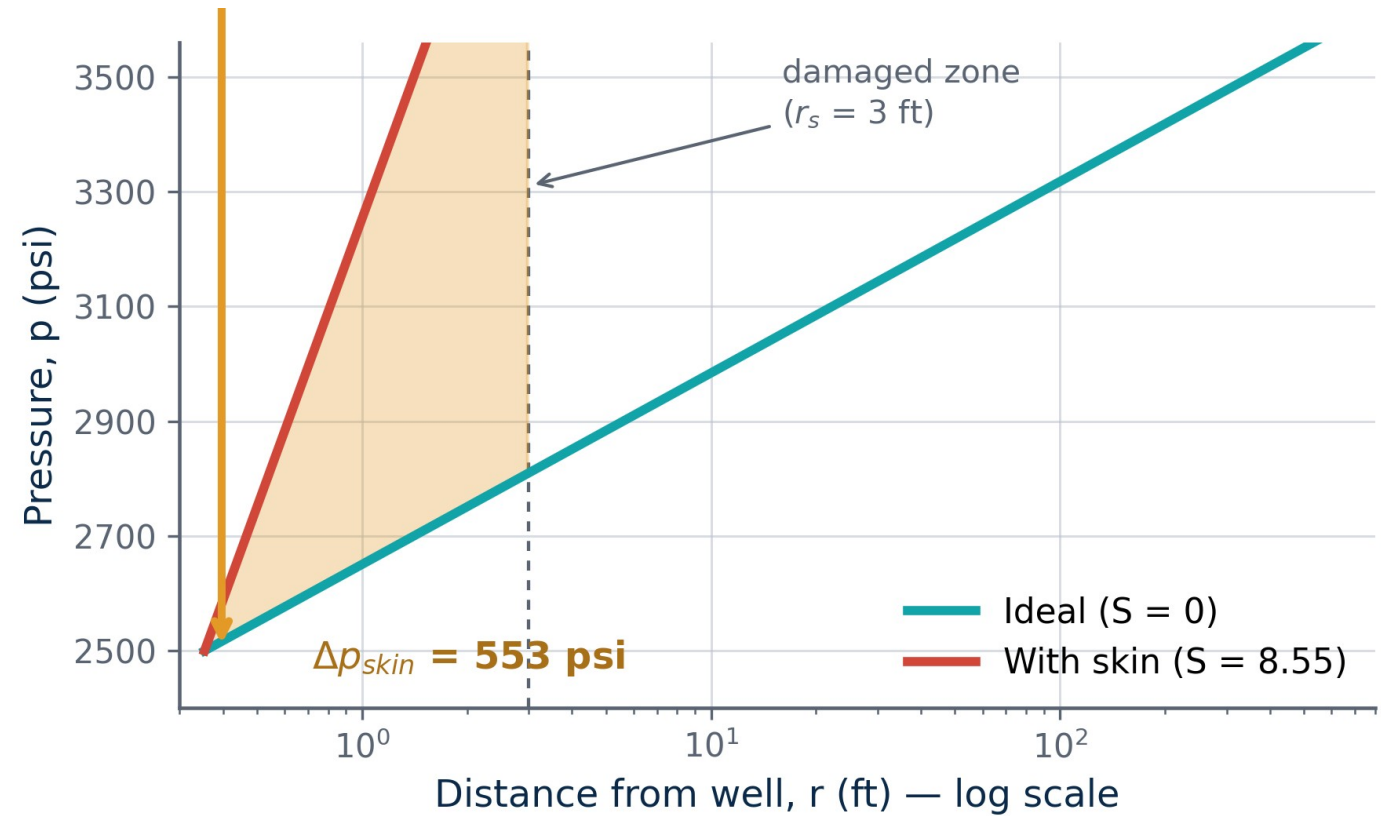
Mechanisms

Quantification

Solutions

Next steps

- $S > 0$: damaged
- $S = 0$: ideal, undisturbed
- $S < 0$: stimulated (acid, fracture)
- Skin converts disturbance into extra pressure drop



$$\Delta p_{skin} = 141.2 q B \mu S / (k h)$$

In the radial-flow equation, skin adds directly to resistance

Fundamentals

Mechanisms

Quantification

Solutions

Next steps

$$q = 0.00708 k h (\bar{p} - p_{wf}) / [\mu B (\ln(r_e/r_w) - 0.75 + S)]$$

q oil rate (STB/d)

k permeability (md)

h net thickness (ft)

$\bar{p}$ average reservoir pressure (psi)

p_{wf} flowing bottomhole pressure (psi)

μ viscosity (cP)

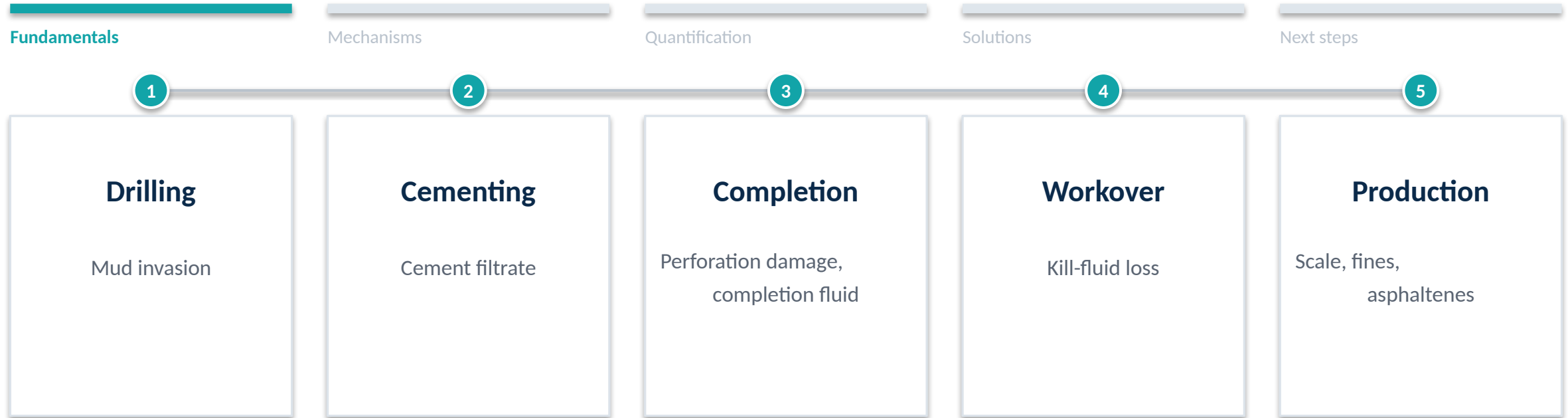
B formation volume factor

r_e, r_w drainage / well radius (ft)

S skin (dimensionless)

- Skin adds to the geometric term
- Here $\ln(r_e/r_w) - 0.75 \approx 6.9$
- So $S \approx 7$ halves the rate

Damage can occur at every stage of the well's life



This talk focuses on drilling-fluid damage

Drilling-fluid invasion is the primary source of damage

Fundamentals

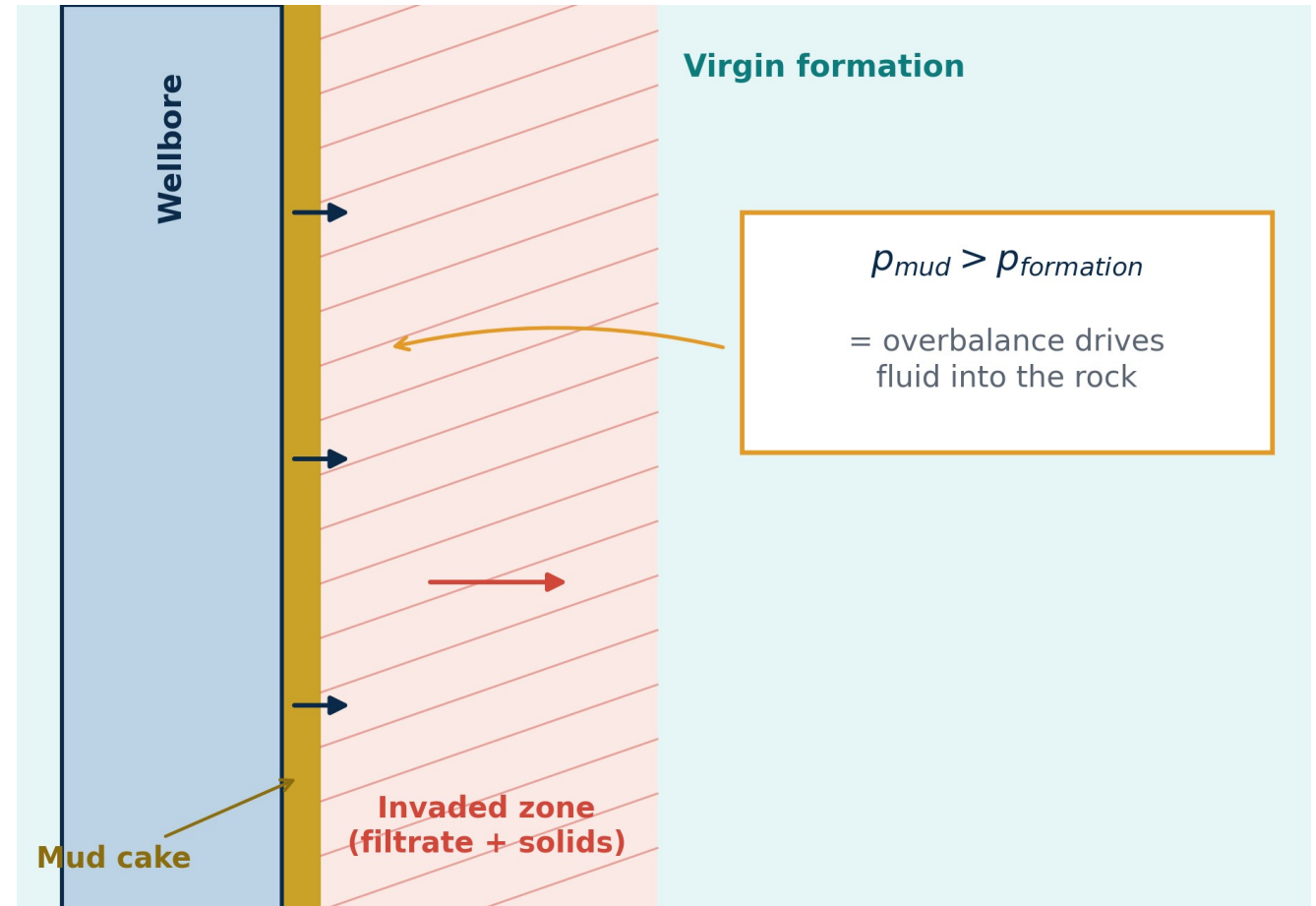
Mechanisms

Quantification

Solutions

Next steps

- Overbalance pushes fluid into the rock
- Spurt loss, then filtrate, then filter cake
- Severity grows with overbalance and exposure time
- Mud properties and permeability also matter



Solids and fines plug pore throats

Fundamentals

Mechanisms

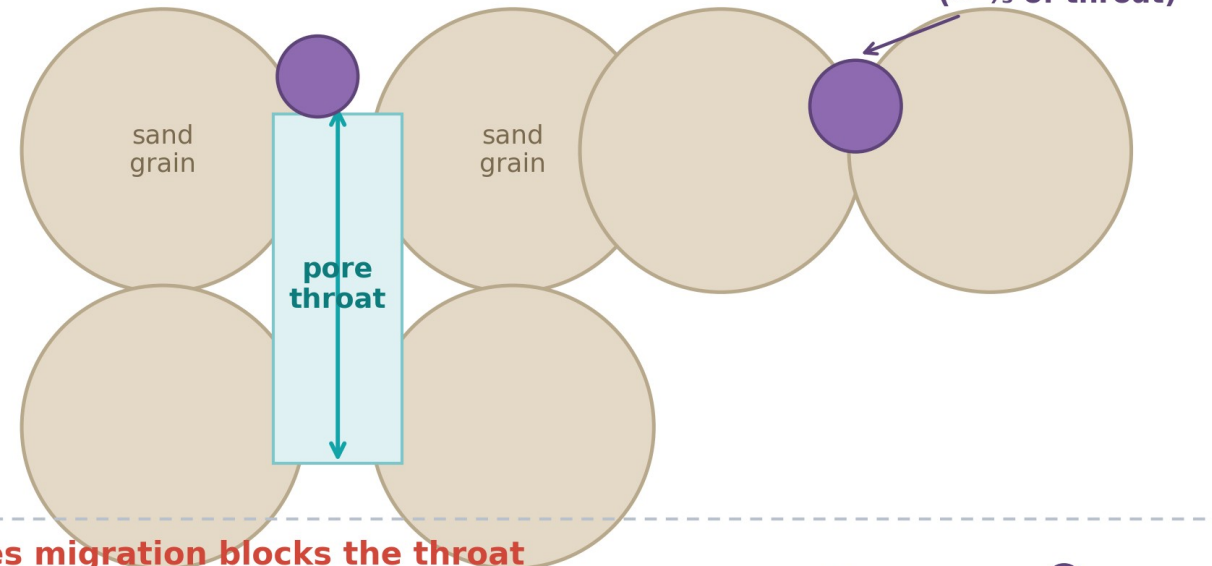
Quantification

Solutions

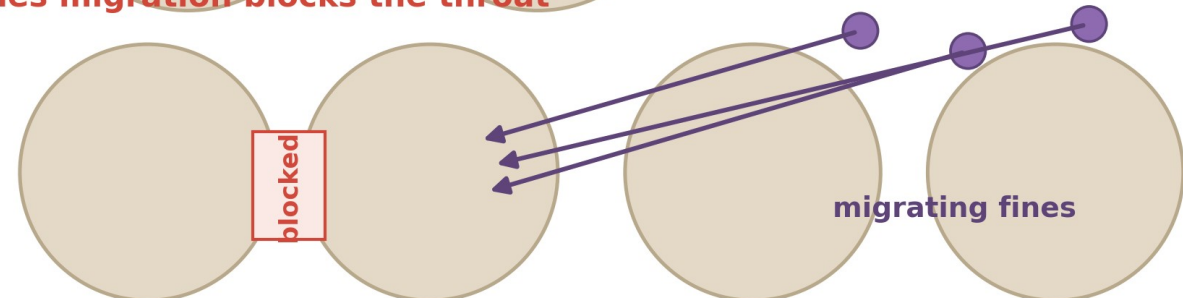
Next steps

- Mud solids bridge or enter pore throats
- Formation fines detach and migrate
- Result: severe, hard-to-reverse permeability loss

Intact pore throat



Fines migration blocks the throat



Incompatible water makes clays swell and migrate

Fundamentals

Mechanisms

Quantification

Solutions

Next steps

Clay	Main damage behaviour
Smectite	Swells, blocks pores
Illite	Migrates as fines
Kaolinite	Detaches and migrates
Chlorite	Iron precipitates with acid

- Fresh filtrate triggers swelling and dispersion
- Salinity and ion composition matter
- Fix: KCl brines, clay stabilizers

Capillary forces trap filtrate near the wellbore

Fundamentals

Mechanisms

Quantification

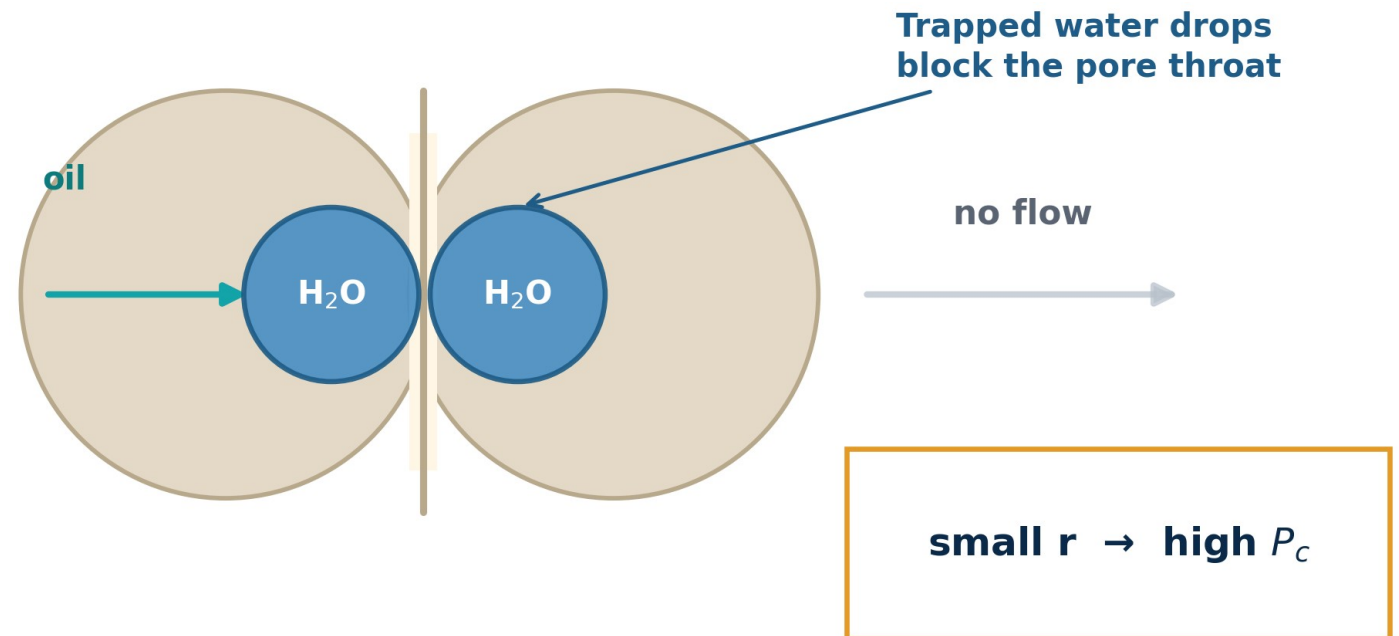
Solutions

Next steps

- Water block cuts hydrocarbon relative permeability
- Strongest in low-permeability rock
- Emulsions and wettability change add resistance

$$P_c = 2\sigma \cos\theta / r$$

(σ = interfacial tension, θ = contact angle, r = pore-throat radius)



Fluid incompatibility creates precipitates in the pores

Fundamentals

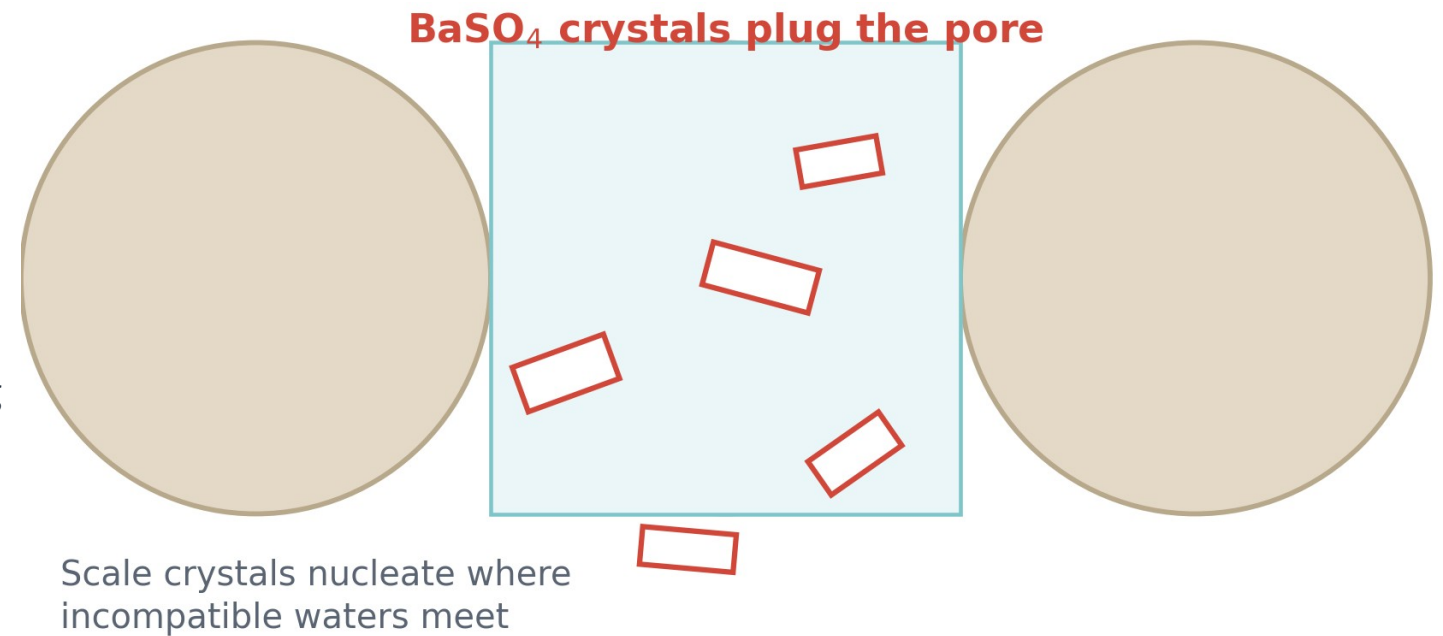
Mechanisms

Quantification

Solutions

Next steps

- Filtrate + brine can form CaCO_3 , CaSO_4 , BaSO_4
- Acids or high-pH fluids: gels, sludges
- Prevent: lab compatibility tests before pumping



Damage is detected by tests and production data

Fundamentals

Mechanisms

Quantification

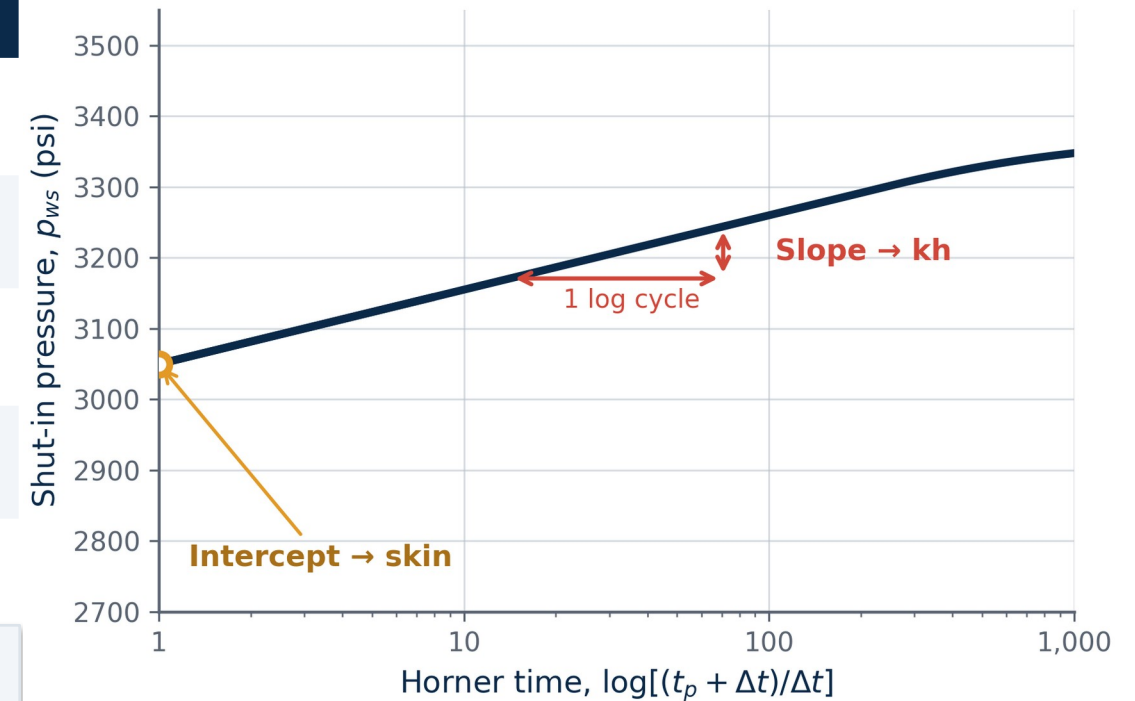
Solutions

Next steps

Method	What it shows
Pressure buildup test	Total skin
PI versus offset wells	Underperformance
Core flow test	Return permeability
Production logging	Inflow profile
Nodal analysis	Expected versus actual rate

$$\text{Return permeability} = k_{\text{after}} / k_{\text{before}}$$

Test skin is total skin, not only damage



Hawkins' equation links the damaged zone to skin

Fundamentals

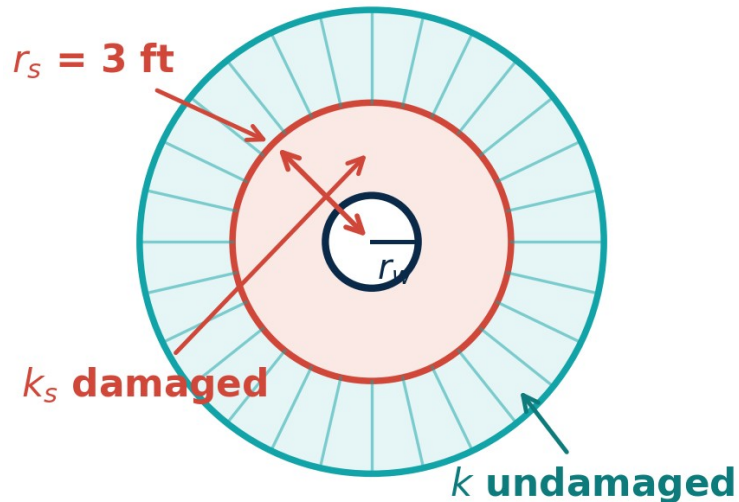
Mechanisms

Quantification

Solutions

Next steps

$$S = (k/k_s - 1) \ln(r_s/r_w)$$



Uniform cylindrical damaged zone

k

undamaged permeability (md)

k_s

damaged permeability (md)

r_s

damage radius (ft)

r_w

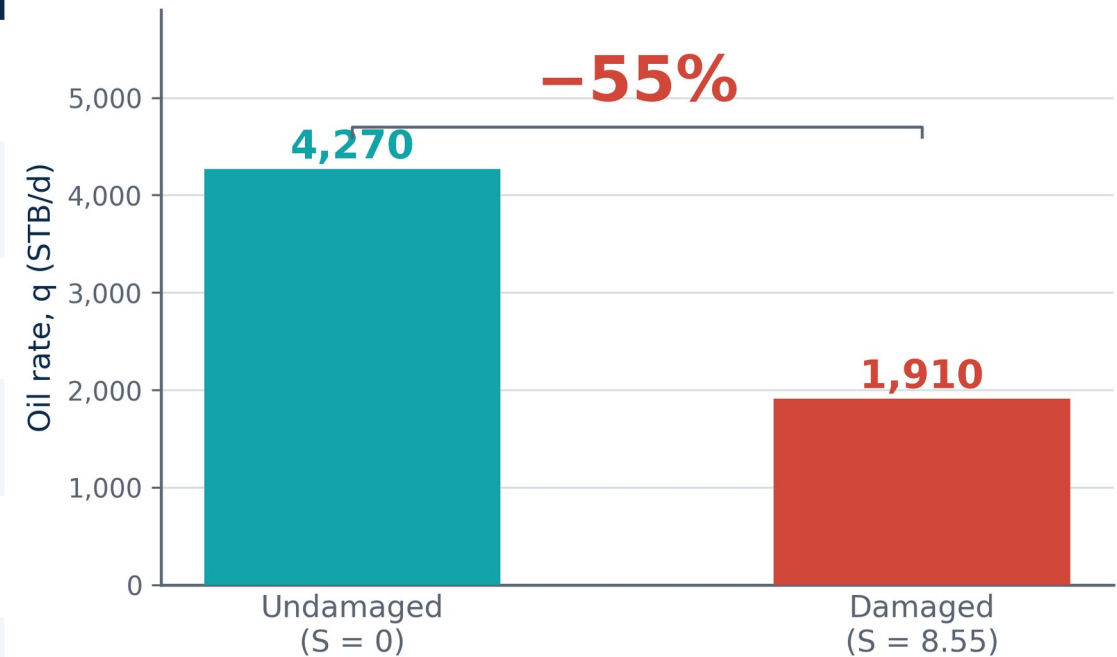
wellbore radius (ft)

- Severity (k/k_s) acts linearly
- Depth (r_s) acts logarithmically
- Assumes a uniform, cylindrical damaged zone
- Well-test skin is total skin

Worked example: a skin of 8.5 costs 55% of productivity

ILLUSTRATIVE DATA

INPUT	OUTPUT
k, k_s : 100, 20 md	$S = 8.55$
$h = 50$ ft	$q(S = 0) = 4,270$ STB/d
r_w, r_e : 0.354, 745 ft	$q(\text{damaged}) = 1,910$ STB/d
$r_s = 3$ ft	$FE = 0.447$
$\bar{p}, p_{wf}$: 3,500, 2,500 psi	$\Delta p_{\text{skin}} = 553$ psi
$\mu = 1$ cP, $B = 1.2$ rb/STB	



Severity of permeability reduction matters more than depth

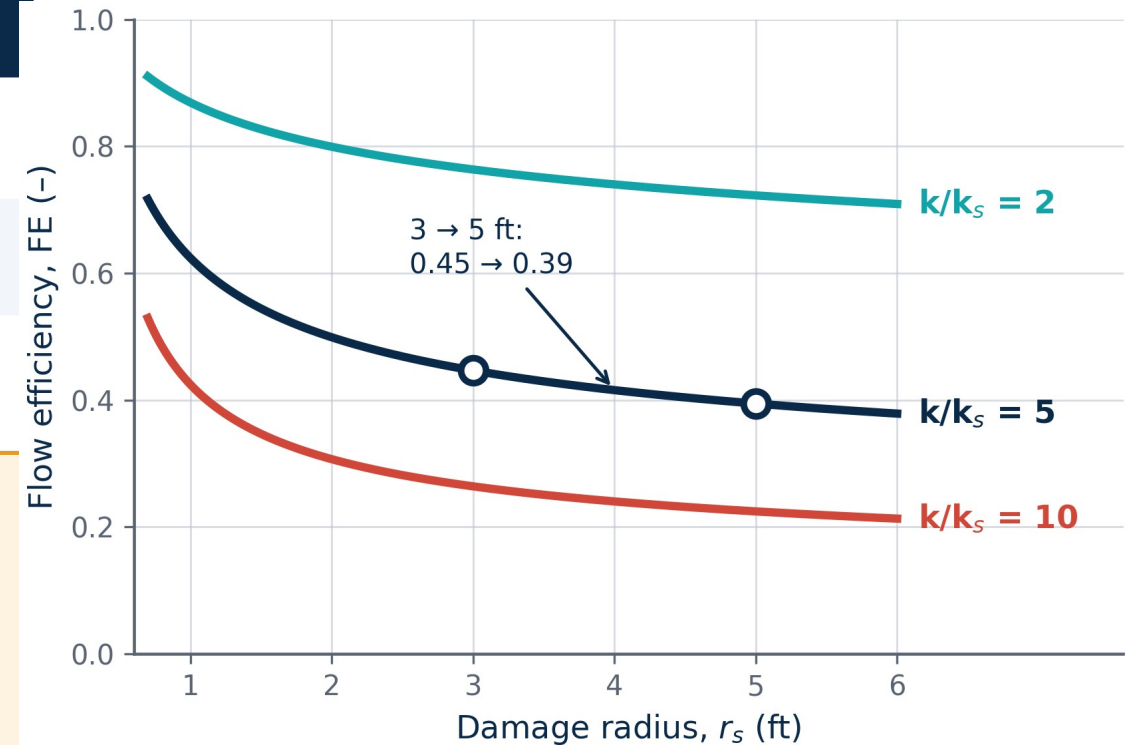
ILLUSTRATIVE DATA

k/k_s	$r_s = 1$ ft	$r_s = 3$ ft	$r_s = 5$ ft
2	1.0 / 0.87	2.1 / 0.76	2.6 / 0.72
5	4.2 / 0.62	8.5 / 0.45	10.6 / 0.39
10	9.3 / 0.42	19.2 / 0.26	23.8 / 0.22

Depth 3 → 5 ft: FE 0.45 → 0.39

Severity 2 → 10: FE 0.76 → 0.26

$S = \text{skin}$ · FE = flow efficiency



Author's calculation. Same reservoir as slide 16.

Prevention starts with fluid design

Fundamentals

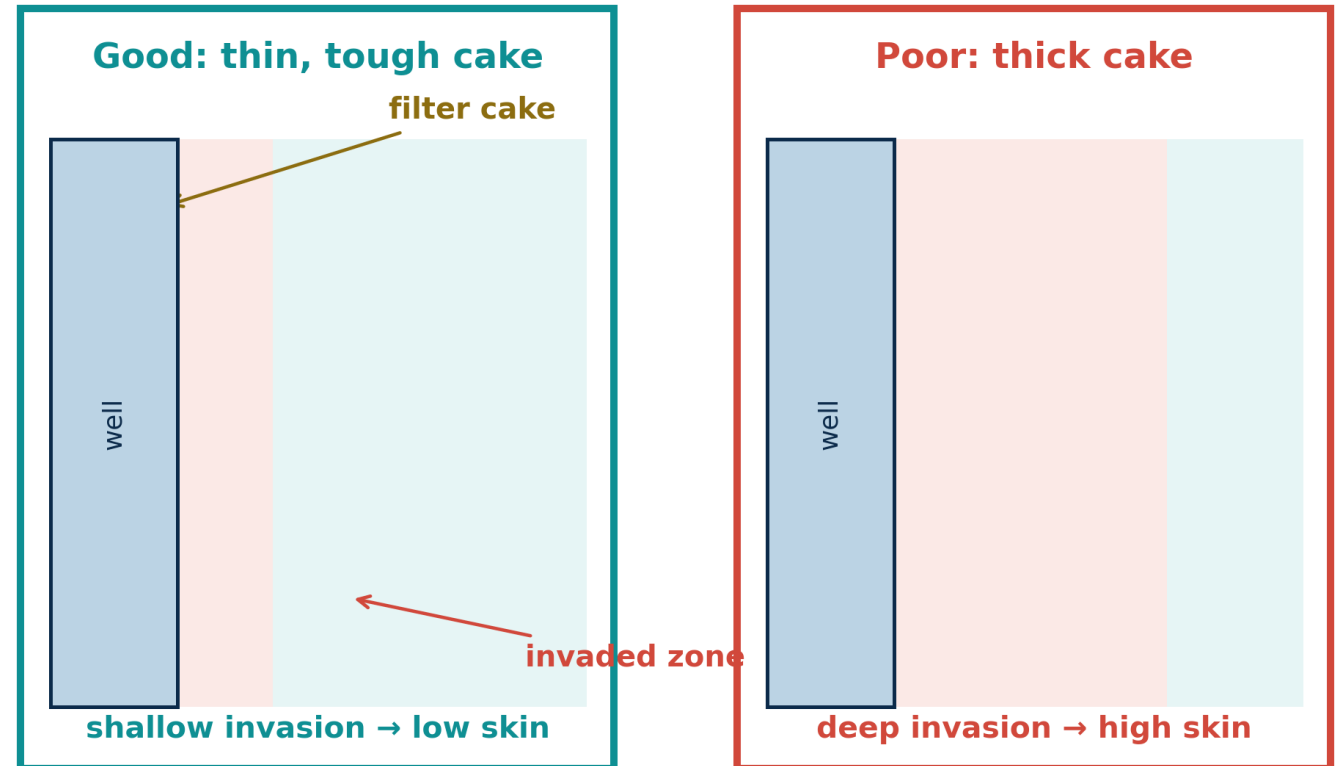
Mechanisms

Quantification

Solutions

Next steps

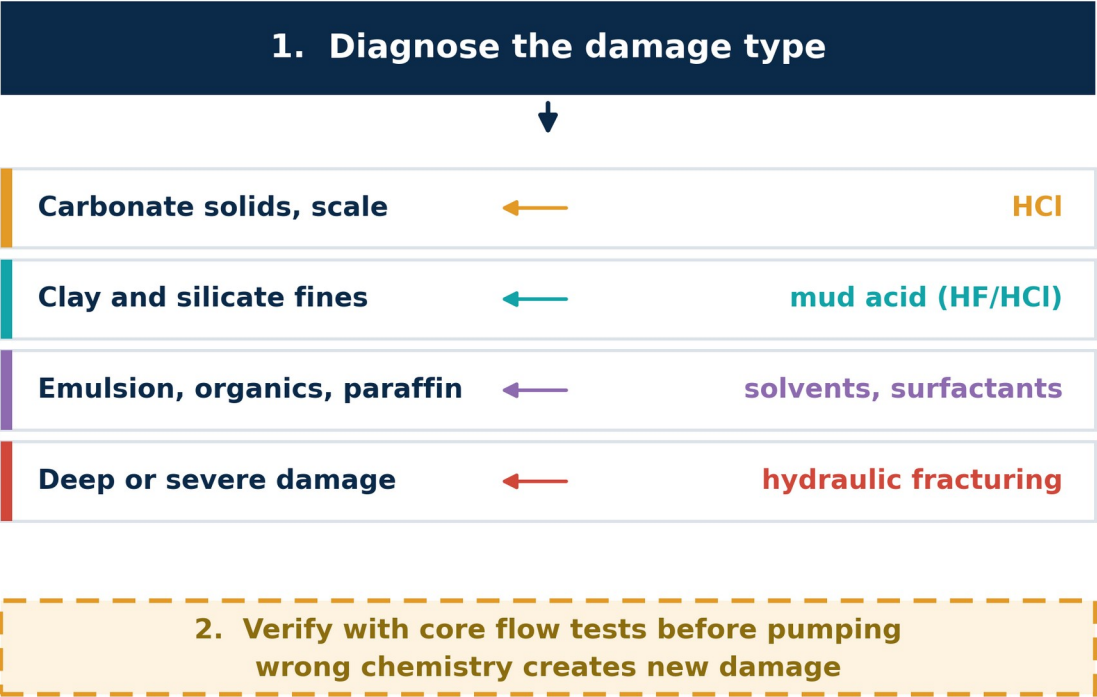
- Size bridging agents $\approx \frac{1}{3}$ of pore throats
- Use acid-soluble CaCO_3 bridging particles
- Low filtrate loss: thin, tough filter cake
- Inhibitive brines (KCl) protect clays
- Minimize overbalance and exposure time



Remediation must match the damage type

Damage	Typical treatment
Carbonate solids, scale	HCl
Clay and silicate fines	Mud acid (HF/HCl)
Emulsion, organics, paraffin	Solvents, surfactants
Deep or severe damage	Hydraulic fracturing

Diagnose first. Wrong treatment creates new damage.

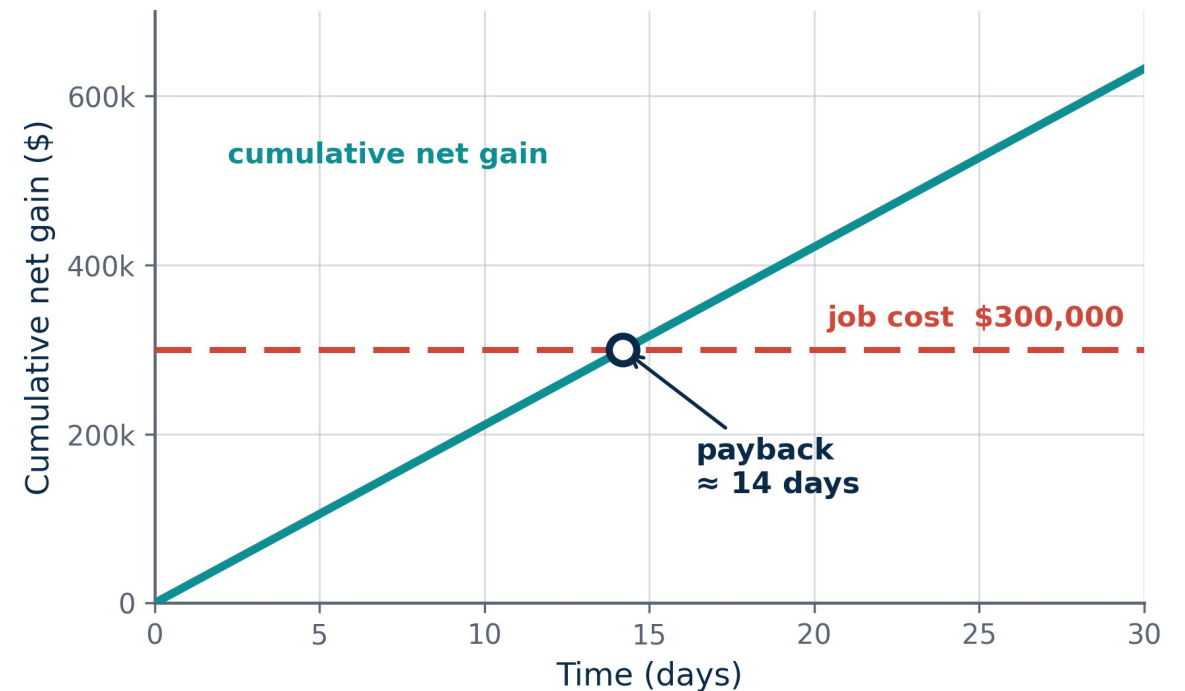


Remediation can pay back fast, but prevention avoids cost and risk

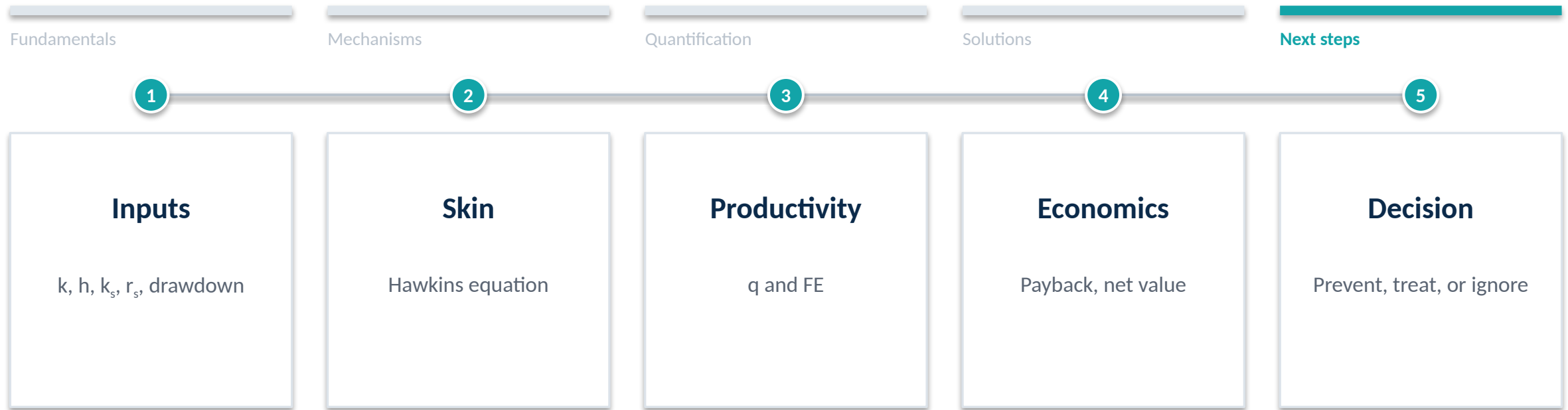
ILLUSTRATIVE DATA

- Acidizing example: S falls from 8.55 to 2
- Rate rises 1,910 $\rightarrow$ 3,310 STB/d
- Payback $\approx$ 14 days (assumed values)
- HSE: acid handling, flowback disposal
- Prevention avoids both cost and risk

Assumed: margin \$30/bbl $\cdot$ 50% of gain sustained
job cost \$300,000 $\cdot$ net gain $\approx$ \$21,075/day



Next step: a skin-based screening model for drilling-fluid damage



Gap: damage is often treated qualitatively

Tools: Excel or Python, literature return-permeability data

Output: sensitivity charts and a decision guide

Conclusions: damage is quantifiable and largely preventable

1. Mechanisms

- Invasion damages by plugging, swelling, trapping

2. Quantification

- Example: skin 8.5 costs 55%
- Severity outweighs depth

3. Strategies

- Prevention beats remediation
- Match treatment to damage type

References and questions

1. Hawkins, M.F. (1956). A note on the skin effect. Trans. AIME, 207.
2. van Everdingen, A.F. (1953). The skin effect and its influence on the productive capacity of a well. Trans. AIME, 198.
3. Krueger, R.F. (1986). An overview of formation damage and well productivity in oilfield operations. JPT, 38(2).
4. Abrams, A. (1977). Mud design to minimize rock impairment due to particle invasion. JPT, 29(5).
5. Earlougher, R.C. (1977). Advances in Well Test Analysis. SPE Monograph 5.
6. Civan, F. Reservoir Formation Damage. Gulf Professional Publishing.
7. Economides, M.J. et al. Petroleum Production Systems. Prentice Hall.
8. Bourgoyne, A.T. et al. (1986). Applied Drilling Engineering. SPE.

Thank you.
Questions?

Verify every reference in your library (volume, pages, editions) before submitting.

Backup: Hawkins' formula comes from series radial resistances

Damaged zone: $\Delta p_s = [q B \mu / (0.00708 k_s h)] \ln(r_s/r_w)$

Undamaged zone: $\Delta p_u = [q B \mu / (0.00708 k h)] [\ln(r_e/r_s) - 0.75]$

Sum: $\ln(r_e/r_w) - 0.75 + (k/k_s - 1) \ln(r_s/r_w)$

Skin term: $S = (k/k_s - 1) \ln(r_s/r_w)$

Both zones are in series, so the pressure drops add — and the extra term is the skin.

Backup: test skin combines several effects

$$S_{\text{total}} = S_{\text{damage}} + S_{\text{perforation}} + S_{\text{partial penetration}} + S_{\text{deviation}} + S_{\text{non-Darcy}}$$

Only S_{damage} is removable by damage treatment

A high test skin is not automatically a formation-damage problem — check completion and geometry first.

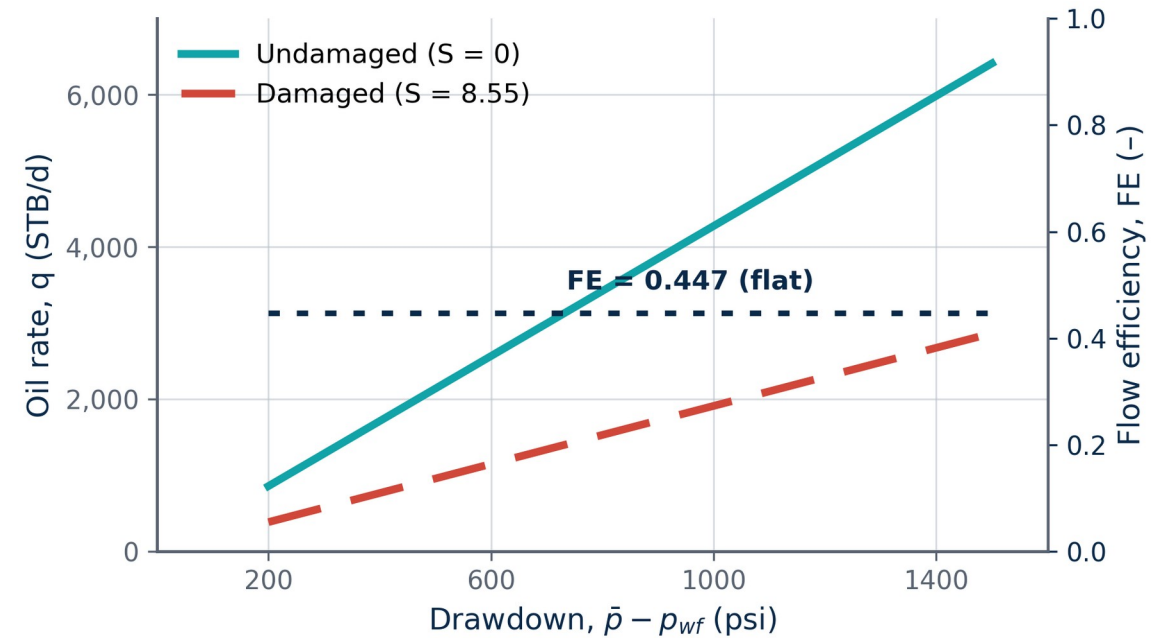
Backup: bridging theory sets particle size

- Median bridging particle $\approx \frac{1}{3}$ median pore throat
- Bridging solids $\geq 5\%$ by volume (Abrams)
- Too fine: deep invasion. Too coarse: no bridge

Sizing is a balance: fine enough to form a bridge, coarse enough to stay on the wall.

Backup: drawdown scales rate, not flow efficiency

- q scales linearly with drawdown
- FE depends only on $\ln(r_e/r_w) - 0.75$ and S
- Relative loss is independent of drawdown



Backup: fluid comparison (qualitative)

Fluid	Main advantage	Main damage risk
Water-based	Low cost	Clay swelling, water block
Oil-based	Shale inhibition	Wettability change, emulsions
Formate brine	Low solids	High cost
Polymer + CaCO_3	Thin, acid-soluble cake	Polymer residue

Check this table against your course notes. The risks depend on the formation.